ALGORITHM FOR AUTOMATED SOLUTION OF ELEMENTARY PHYSICS PROBLEMS (7TH GRADE)

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Abstract

The proposed table of physical quantities allows replacing the fragmented list of formulas with a unified concept. The table reflects all necessary connections between physical quantities and their symmetric relationships. The structure of the table facilitates the memorization of physical laws and allows for the convenient application of various algebraic tools (real and imaginary numbers, orthogonal functions, matrices, quaternions, tensors, differential forms, systems of differential equations) for solving specific problems.

Introduction

Historically, tables have served as the foundation of education, from the multiplication table to Mendeleev's Periodic Table. In physics, a Table of Physical Formulas (TPF) can serve as a base for systematic categorization.
Since 1822 (J.B.J. Fourier), many scientists have contributed to the development of measurement operations: J. Maxwell (1881), A.U. Rucker (1888), Lord Kelvin (1889), E. Buckingham (1914), R. Tolman (1917), A. Eddington (1928), G. Burniston Brown (1935), Robert Oros di Bartini (1950).
As physics uses mathematics as a language, it is necessary to mention the mathematicians who laid the foundations of four-dimensional space-time: B. Riemann, W. Clifford, H. Hopf, H. Lorentz, H. Poincaré. Gravity theory was further developed by D. Hilbert, A. Einstein, T. Kaluza, and R.O. di Bartini. In modern physics, these ideas continue in "Theories of Everything" based on exceptional simple groups and 3-brane theory.
In school physics education, it is essential to lay the groundwork for understanding modern physical theories. As Sir A. Eddington remarked: "The essence of all exact sciences consists in taking readings of pointers...". Today, this function is performed by digital sensors, and data visualization is carried out through scripts and virtual instruments.

Table of Physical Formulas

We present a table in which the main formulas of general physics for 7th-8th grade secondary school curricula are systematized in a concise graphical form. Physical quantities and their connections serve as reference points for memorizing the material.

Mathematical Models

The 7th-grade curriculum relies on geometric concepts: length, area, and volume. For instance, the volume of a solid parallelepiped is equal to the product of its three sides. The algorithm for calculating it can be divided into two steps: calculating the area (multiplying length by width) and

subsequently multiplying by height.
Another mathematical model is based on calculating the area of a circle and the volume of a sphere. It is advisable to explain to students the sequence: circumference (2*pi*r) → area of a circle (pi*r^2), where the area of the circle is the antiderivative of the circumference.
It is useful to introduce the concept of a Solid Angle (polyhedral or 3D angles) and the unit of measurement, the steradian. A sphere forms a solid angle equal to 4*pi steradians (a full solid angle).
For the transition from the area of a circle (pi*r^2) to the surface area of a sphere (4*pi*r^2), see the section below: "Digression on the area and surface of a torus". This material can serve as a base for studying additional chapters on quaternion mathematics in higher grades.
Step two: Surface area of a sphere (4*pi*r^2) → volume of a sphere ((4/3)*pi*r^3). Here, the volume of the sphere is the antiderivative of the surface area of the sphere.
These mathematical models can be used for the automated solution of general physics problems for the 7th grade.

Digression on the area and surface of a torus:

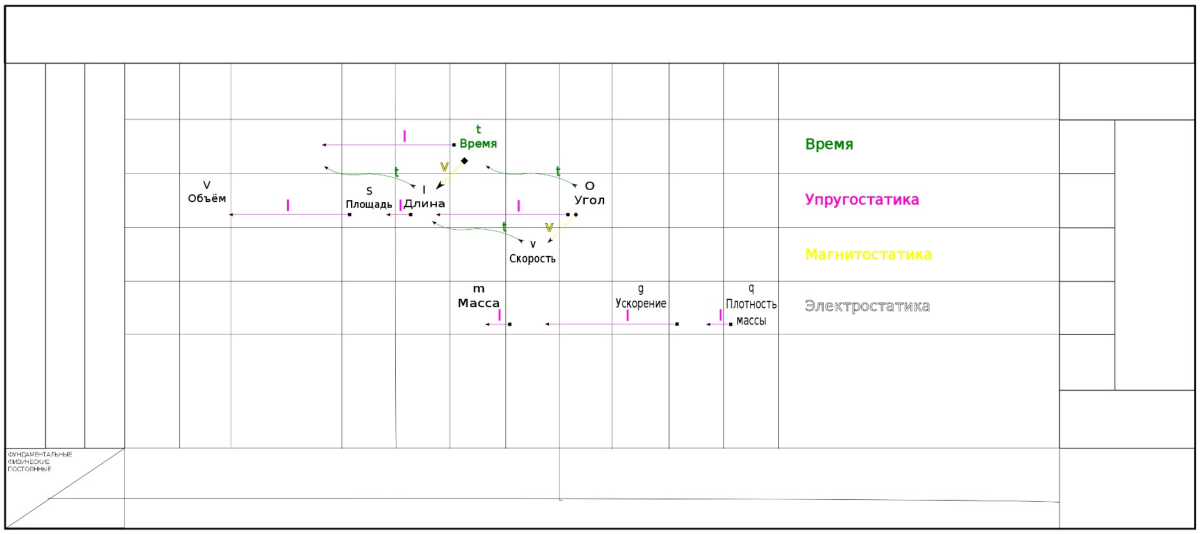
The surface area of a hypersphere is equal to the volume of a torus of one lower dimension.
`Sn + 1 / Vn = 2πR`
And this formula Vn * 2πR can be understood as the volume of a hypertorus. Suppose Vn = 2R, where n = 1. Then V_tor = S1 + 1 = 4 π R^2. However, according to another formula Vn / Sn − 1 = R / n, at n = 3, we obtain the volume of a sphere V3 = 4/3 π R^3.

Table of Physical Formulas (Analysis)

For the automated solution of general physics problems, it is advisable to establish relationships between physical quantities. If:

- l – extension (length)
- S – area
- V – volume

Then the mathematical model mentioned above can be written as follows:
`V ← S ← l`
Where ← denotes the operation of finding the antiderivative with respect to spatial extension. In the case of a parallelepiped, this is the sequential multiplication of length (l) by width (arrow directed to S) and height (arrow directed to V).
If the solid angle is denoted by the capital omega (O), then our graph can be extended:
`V ← S ← l ← O`
We draw this graph in the table. The arrows (Operation of partial derivative with respect to space) are displayed in magenta. The derivative is taken against the direction of the arrow. For example, the derivative of the function for the area of a circle, which depends only on the radius of the circle, is the function for the circumference (perimeter of the circle), which also depends only on the radius.



A solid angle can be expressed through plane angles. A plane angle is a dimensionless quantity—that is, a quantity which, when multiplied by any physical parameter, assumes its dimension. Consider, for example, a mechanical clock; there are mechanical clocks with a dial for a full day. By the angle of displacement of the hour hand, it is possible to measure time intervals, which are usually represented by the length of the arc of the dial's circle. The arc length depends on the radius of the circle or the length of the hour hand. Thus, we can write the relationship:

`t ← O`

Where:

- t – time
- O – angle
- ← denotes the operation of finding the antiderivative with respect to time.

Here, the antiderivative of angle O with respect to time is used, since angle O is a time parameter in our clock model. Applying a similar operation twice to length (in physics, this operation is called the partial derivative with respect to time), we obtain the following graph:

`l ← v ← g`

Where:

- l – the path length covered by a material point
- v – the velocity of a material point at each moment in time
- g – the acceleration of a material point

We display these relationships in the table in the form of winding green arrows. It can be noted that the relationships between physical quantities are obtained by translation (vertical and horizontal shifting) of the corresponding groups of arrows (time, space, velocity), but it must be taken into account that physical quantities are of type one (conventionally distributed) and type two (concentrated).

Physical Quantities of Type Two

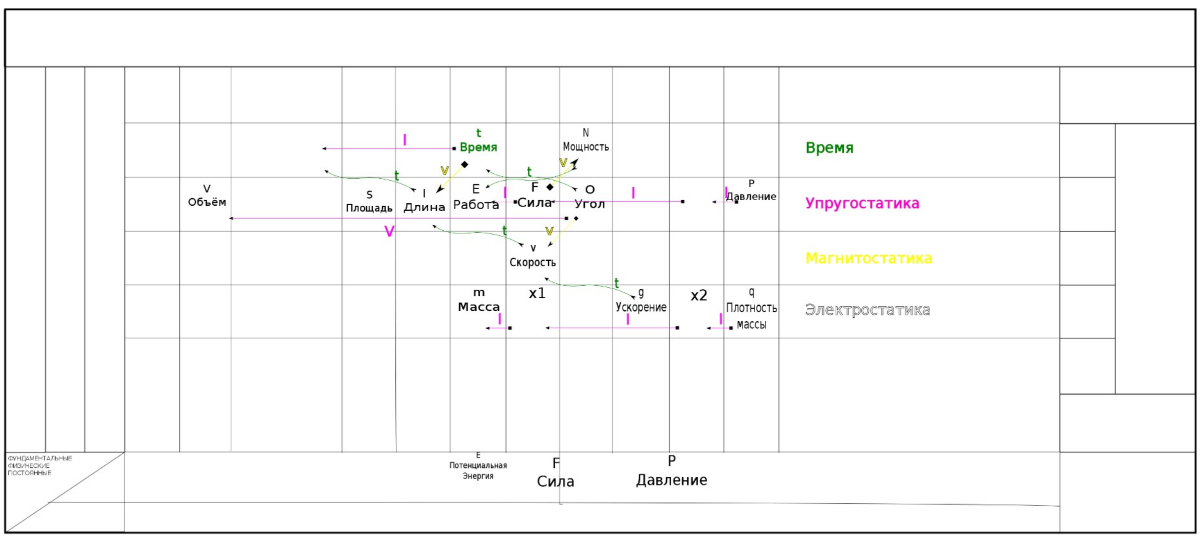
The physical quantities of the second type studied in the 7th grade include: Mass, Density, Force, Pressure, Energy, Power, as well as Work.

`m ← x1 ← x2 ← q`

Where:

- m – mass
- x1 – linear density
- x2 – surface density
- q – density

These relationships are also displayed in Figure 1. All these physical quantities belong to the second type.



When dividing mass by density, we obtain volume. In a non-algorithmic way, this can be done as follows: first, divide the mass by the mass contained within the density. This gives a dimensionless quantity O, which indicates how many elementary masses (or body parts) are contained in the required value. Then, this number must be multiplied by the volume of the elementary part, using a simple linear model of the body's volume as the product of length, height, and width.

Thus, we see the connection between mass—a type two quantity—and volume—a type one quantity.

Similarly, mass is obtained from density by simple integration over the volume. This fact is also part of the Gauss-Ostrogradsky theorem, if one considers that density q is a 3-form and, by definition, is the divergence of the 2-form x2.